

Multiple Choice (10 marks)

1. Determine whether the polynomial in $\mathbb{Z}[x]$ satisfies an Eisenstein criterion for irreducibility over $\mathbb{Q}$. $8x^3 + 6x^2 - 9x + 24$
 - (a) Yes, with $p=2$.
 - (b) Yes, with $p=3$.
 - (c) Yes, with $p=5$.
 - (d) No.
2. If a polynomial $f(x)$ over the real numbers has the complex numbers $2 + i$ and $1 - i$ as roots, then $f(x)$ could be
 - (a) $x^3 + 5x^2 + 4x + 1$
 - (b) $x^4 - 6x^3 + 15x^2 - 18x + 10$
 - (c) $x^3 - x^2 + 4x + 1$
 - (d) $x^4 + 7x^2 + 10$
3. Statement 1 | Every integral domain with characteristic 0 is infinite. Statement 2 | Every integral domain with prime characteristic is finite.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
4. Let $I \neq A \neq -I$, where I is the identity matrix and A is a real 2×2 matrix. If $A = A(-1)$, then the trace of A is
 - (a) 2
 - (b) 1
 - (c) 0
 - (d) -1
5. Let $f(x,y) = x^2 - 2xy + y^3$ for all real x and y . Which of the following is true?
 - (a) f has all of its relative extrema on the line $x = y$.
 - (b) f has all of its relative extrema on the parabola $x = y^2$.
 - (c) f has a relative minimum at $(0, 0)$.
 - (d) f has an absolute minimum at $(2/3, 2/3)$.

True / False (10 marks)

Answer True or False

6. True or False: The probability that 0 will not be chosen when k digits are picked from $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ is $(9/10)^k$.
7. True or False: The unity of a subring must always be the same as the unity of the ring it is contained in.
8. True or False: The polynomial $x^2 + 1$ has two distinct zeros in the finite field $\mathbb{Z}_2$.
9. True or False: The probability that the player who makes the first toss wins the game is $1/4$.
10. True or False: In a group, the equation $(ab)^n = a^n b^n$ holds true for all integers n .

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. The acute angles of a particular isosceles trapezoid each measure 40 degrees. What is the measure, in degrees, of each obtuse angle in the trapezoid?
12. Analyze the process of determining the units digit of 7 raised to the power of 25. Discuss any patterns observed in the units digits of powers of 7 and explain your reasoning.

End of Paper